

Project ID:
R26-IT-114

1. Topic (12 words max)

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	No
Year	No

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Neurodevelopmental learning disorders affect millions of children worldwide and result from cognitive, neurological, and environmental factors [1]. These disorders cause persistent difficulties in acquiring and applying academic skills such as reading, writing, mathematical reasoning, and memory, which negatively affect children's learning and long-term educational outcomes [1]. Common conditions include dyslexia, dyscalculia, dysgraphia, and working memory deficits [1][3].

Although educational and therapeutic interventions are available, many existing approaches suffer from limitations related to accessibility, personalization, and early identification. A significant number of children remain undiagnosed or receive delayed support due to shortages of trained specialists, high assessment costs, and limited access to individualized learning programs, particularly in underserved regions [1]. Consequently, affected children often experience academic challenges, reduced self-confidence, and emotional distress [1].

This issue is critical because untreated learning disorders can negatively impact academic achievement, social integration, and future employment opportunities, thereby reducing quality of life and societal productivity [2]. Research has shown that working memory plays a key role in academic attainment, emphasizing the importance of early and effective intervention to minimize learning gaps and support positive developmental outcomes [2]. However, current support methods such as special education programs and therapist-guided activities often rely on manual monitoring and lack adaptive mechanisms tailored to individual learning needs [1].

To address these challenges, this research proposes a rule-based, game-based mobile learning platform for children aged 6–8 years. The system evaluates learner performance using predefined rules and dynamically adjusts learning activities based on progress, while providing progress dashboards for parents and teachers to enable monitoring and timely intervention.

This research falls under the FoC Research Cluster: Technology Integration & Management and aligns with the EdTech and Digital Health domains by applying artificial intelligence concepts to educational support systems. The proposed solution contributes to inclusive education (SDG 4), supports children’s cognitive and emotional well-being (SDG 3), and improves access to learning support for underserved communities (SDG 10).

Reference

1. KEC Intriago, LMA Rodríguez... - ... Research Journal of ..., 2021 - pdfs.semanticscholar.org. [PDF] . Available : [Specific learning difficulty: autism, dyscalculia, dyslexia and dysgraphia](#)
2. Alloway, T. P., & Alloway, R. G. (2010). Investigating the predictive roles of working memory and IQ in academic attainment. *Journal of Experimental Child Psychology*, 106(1), 20–29. [Online] . Available : <https://www.sciencedirect.com/science/article/abs/pii/S0022096509002021?via%3DiHub>
3. Learning Disabilities Association of America, *What is Dyscalculia?* Available at: <https://ldaamerica.org/what-is-dyscalculia/>

6. Brief description of Existing Research and Systems (300 words max)

Existing solutions for supporting children with neurodevelopmental learning disorders such as dyslexia, dyscalculia, dysgraphia, and memory deficits mainly rely on traditional educational and clinical approaches . These approaches include teacher-led remedial instruction, special education classroom programs, and therapist-guided learning sessions [1]. Although such methods can be effective, they often require significant time, trained professionals, and physical access to learning centers, making them less accessible for children living in rural or underserved areas [2].

In recent years, several mobile and computer-based learning applications have been developed to support children with learning difficulties [3]. Many of these applications focus on specific skills such as reading, basic mathematics, or handwriting practice, and are generally designed for mainstream learning rather than targeted intervention for neurodevelopmental learning disorders . Some platforms use simple game elements to improve learner engagement and provide basic progress reports for parents or teachers [4].

However, most existing systems lack adaptive mechanisms that personalize learning based on individual performance and learning pace [3]. In addition, many solutions address only a single learning disorder and fail to offer integrated support for children experiencing multiple learning challenges .

Previous research highlights that game-based learning and adaptive educational systems can enhance motivation, engagement, and learning outcomes among children with learning disorders [4]. Adaptive learning techniques allow instructional content to be adjusted according to learner performance, supporting more effective intervention [3]. Despite these benefits, many existing studies focus on individual disorders or small-scale experimental environments, limiting their applicability in real-world educational contexts [4]. Furthermore, continuous progress monitoring and accessible feedback for parents and educators are often limited [2]. These gaps indicate the need for a simple, adaptive, and accessible therapeutic mobile learning platform that provides integrated and personalized support for children with neurodevelopmental learning disorders.

Reference

1. **American Psychiatric Association**, *Diagnostic and Statistical Manual of Mental Disorders (DSM-5)*, 5th ed. Washington, DC, USA: APA, 2013.
Available at: <https://www.psychiatry.org/psychiatrists/practice/dsm>
2. **World Health Organization**, *Developmental Difficulties in Childhood*, WHO Press, Geneva, Switzerland, 2012.
Available at: <https://www.who.int/publications>
3. **D. Holmes, L. Gathercole, and C. Dunning**, "Adaptive training leads to sustained enhancement of poor working memory in children," *Developmental Science*, vol. 12, no. 4, pp. F9–F15, 2009.
Available at: <https://onlinelibrary.wiley.com/doi/10.1111/j.1467-7687.2009.00848.x>
4. **J. Hamari, J. Koivisto, and H. Sarsa**, "Does gamification work? A literature review of empirical studies," *Proceedings of the 47th Hawaii International Conference on System Sciences*, pp. 3025–3034, 2014.
Available at: <https://ieeexplore.ieee.org/document/6758978>

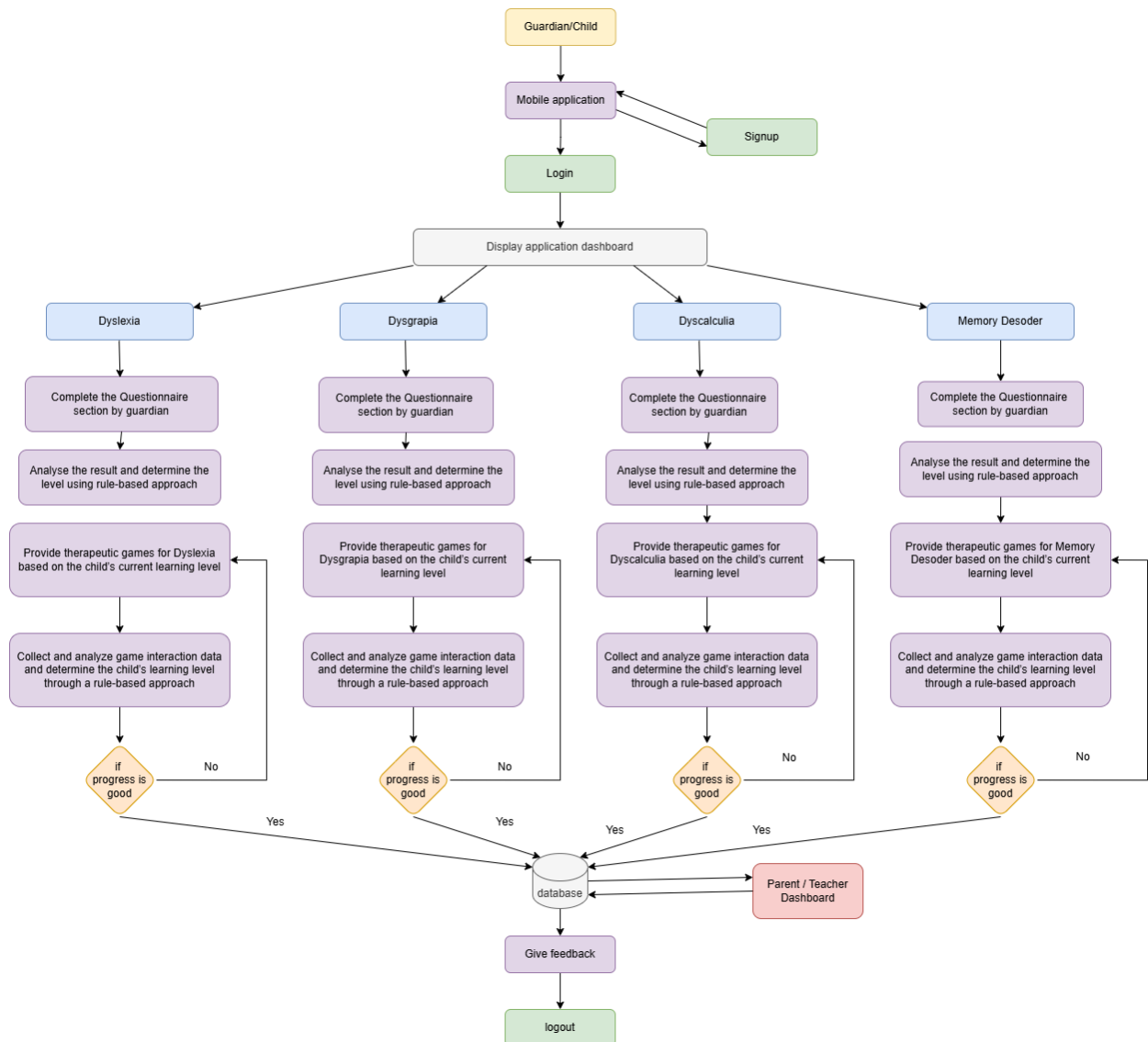
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an adaptive, rule-based, game-based mobile learning platform designed to support children with neurodevelopmental learning disorders such as dyslexia, dyscalculia, dysgraphia, and working memory deficits through personalized and performance-based interventions. The platform enables early screening, individualized exercise assignment, and continuous progress monitoring within a child-friendly and accessible mobile application.

The system follows a closed-loop adaptive learning process that begins with an interactive dashboard used by parents or guardians to initiate the screening process and select relevant learning categories. As an initial step, the system presents a short set of 10 structured screening questions to parents or guardians. These questions gather background information related to the child's learning behavior, attention, memory, academic challenges, and daily learning difficulties. This input provides important contextual information before direct child interaction begins.

Following the parent or guardian questionnaire, the child completes a set of short, age-appropriate game-based tasks related to reading, writing, numerical reasoning, and memory skills. These games are designed to be engaging while capturing key behavioral indicators such as accuracy, response time, error patterns, and task completion behavior. Based on the combined results of the parent questionnaire and the child's game performance, the system applies a rule-based classification approach to identify the child's initial difficulty level as mild, moderate, or high. This level acts as a preliminary screening outcome and helps establish the child's baseline learning profile.

Once the initial level is identified, the system applies rule-based performance evaluation to analyze detailed indicators collected during gameplay. A diagnostic performance summary is generated, which refines the baseline learning profile. Using this profile, the adaptive learning engine dynamically assigns personalized game-based activities targeting the child's specific needs. These include reading exercises for dyslexia, number and logic games for dyscalculia,



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Developing the proposed adaptive, game-based mobile learning platform requires the integration of expertise from education, psychology, and computer science to ensure an effective and reliable solution for supporting children with neurodevelopmental learning disorders. Educational psychologists and special education professionals contribute essential domain knowledge on conditions such as dyslexia, dyscalculia, dysgraphia, and memory deficits. Their expertise ensures that learning activities, assessment methods, and intervention strategies are aligned with established educational and therapeutic standards.

Team members can acquire the technical knowledge and skills required for the project through a combination of structured learning, hands-on practice, collaboration, and continuous improvement. Foundational skills in mobile app development, backend systems, databases, and rule-based system design can be developed through online courses and official documentation. Practical experience is gained by building prototypes, implementing rule-based and adaptive game logic, and experimenting with gamification features. Collaboration through role-based learning, peer discussions, and version control tools enhances technical competence. Additionally, reviewing domain-specific literature and ethical guidelines ensures responsible system design. Continuous testing, feedback, and reflection further refine skills and support successful project implementation.

Relevant educational and behavioral data is essential for designing, testing, and validating the rule-based logic used in this project. The system utilizes baseline assessment results and anonymized interaction data collected from game-based learning activities, including performance measures such as accuracy, response time, error patterns, and learning progression. In addition, qualitative feedback from parents, teachers, and therapists is gathered to better understand each child's learning behavior and to improve personalization and system effectiveness.

The data required for the project is collected using controlled and ethical methods. During the early development stage, publicly available educational datasets, anonymized sample data, and simulated user interactions are used to design and test the system. During system testing and evaluation, real user interaction data is collected only with informed consent. All collected data is anonymized, securely stored, and used solely for research and system improvement, ensuring privacy and confidentiality.

This project requires ethical clearance as it involves children aged 6–8 years with neurodevelopmental learning disorders. Data will be collected from approximately 100 children at Lady Ridgeway Hospital for Children (LRH), Colombo. Approval will be obtained from the relevant institutional ethics review committee before data collection begins. Informed consent will be obtained from parents or legal guardians, and assent from children where appropriate. The study will follow strict ethical principles, including confidentiality, data privacy, voluntary participation, and the right to withdraw at any time. No personally identifiable information will be disclosed or misused.

9. Objectives and Novelty

Main Objective To develop an adaptive, game-based mobile learning platform leveraging artificial intelligence (AI) to support children with neurodevelopmental learning disorders, including dyslexia, dyscalculia, dysgraphia, and memory deficits, by providing personalized educational and therapeutic interventions, tracking learning progress, and enhancing academic skills, engagement, and self-confidence.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22062574	Develop an adaptive learning and screening module for Dyscalculia.	Track children's understanding of numbers, counting, sequencing, place value, and basic arithmetic operations. Collect and evaluate interaction data from digital exercises, identifying specific difficulties for each child. Provide personalized interventions Based on the analysis, the system recommends and delivers adaptive, game-based math exercises tailored to the child's learning needs. Monitor progress and improvements	The novelty of this Dyscalculia module lies in its adaptive, game-based learning approach that goes beyond traditional static exercises. Unlike existing apps, it begins with an initial assessment to determine each child's current math level across multiple skills, including number sense, sequencing, and basic operations. Based on this, the system delivers personalized games tailored to the child's specific difficulties and automatically adjusts difficulty or generates new exercises if improvement is not observed. The platform continuously tracks

		Continuously track performance, adjust exercise difficulty dynamically, and provide visual dashboards for parents and teachers to monitor learning growth.	performance, error patterns, and engagement, providing dashboards for parents and teachers to monitor progress.
IT22128836	Develop game-based dysgraphia intervention module that improves handwriting, spelling, and written expression in children by providing personalized practice within a mobile learning platform.	Track children's writing abilities by evaluating letter formation, spacing, alignment, capitalization, and spelling. Collect and evaluate handwriting patterns, identifying specific difficulties for each child. Based on the analysis, the system recommends and delivers adaptive, game-based writing exercises tailored to each child's needs. Continuously track handwriting performance, updating exercises dynamically and providing visual feedback on learning progress.	Most existing solutions primarily focus on repetitive handwriting practice or simple letter-tracing activities, with minimal assessment of a child's actual writing ability. A major challenge is accurately identifying individual handwriting difficulties such as poor letter formation, spacing, and alignment. This research proposes a game-based mobile application that assesses children's handwriting skills, identifies dysgraphia-related difficulties, and delivers adaptive writing games based on each child's current ability level. The system continuously adjusts game difficulty according to the child's progress and generates reports for parents and teachers.

IT22252418	Dyslexia (Reading) Develop an adaptive mobile learning and intervention module for children with Dyslexia that improves reading accuracy, fluency, phonological awareness, and comprehension through personalized digital activities.	Track children’s reading and spelling abilities by evaluating letter–sound correspondence, phonemic awareness, word decoding accuracy, reading fluency, and spelling consistency. Collect and analyze reading and spelling interaction patterns, identifying dyslexia-specific difficulties such as letter reversals, sound substitutions, omissions, slow decoding, and inaccurate word recognition for each child. Based on the analysis, the system recommends and delivers adaptive, game-based literacy exercises tailored to each child’s needs, focusing on phonics reinforcement, syllable segmentation, word blending, sight-word practice, and spelling correction. Continuously monitor reading and spelling performance by tracking accuracy, response time, error frequency, and comprehension outcomes. The system dynamically adjusts exercise difficulty and provides visual feedback and progress indicators	Existing dyslexia learning applications mainly rely on a single assessment method, such as fixed questionnaires or standard tests, to identify a child’s learning level. These systems usually provide the same learning flow to all children and do not adjust difficulty levels continuously based on the child’s performance. In contrast, the proposed system first identifies the child’s existing dyslexia level using a combined approach that includes a questionnaire completed by parents and short interactive games played by the child. Based on the child’s real-time performance, the app automatically increases or decreases the difficulty level, ensuring that learning activities match the child’s current ability. Unlike existing systems that offer limited feedback to parents, the proposed system also provides a dedicated parent dashboard that displays the child’s daily interactions, progress, and improvement in an easy-to-understand format. This approach supports personalized learning and
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		to illustrate learning improvement and maintain learner engagement.	enables early, continuous parental involvement to improve the child's learning outcomes.
IT22502360	Support and improvement of working memory skills in children with working memory disorder	Use simple games to test how well the child remembers numbers, pictures, words, and sequences. Check correct answers, time taken, and mistakes to understand the child's memory level. Provide suitable game-based activities like matching, recall, and pattern games based on the child's results. Increase or decrease game difficulty based on the child's performance. Monitor memory improvement over time and show progress using simple charts, levels, and feedback. Let parents and teachers view the child's progress and performance through a dashboard.	Most existing memory apps use fixed-difficulty games where all children play the same tasks. However, children with working memory problems have different ability levels and need personalized support. The system analyzes each child's performance in real time while they play memory-based learning games. Using factors such as accuracy, response time, and behavior patterns, the system automatically adjusts memory load, task speed, and game type to suit the child's current ability. In addition, the system identifies specific working memory difficulties such as attention loss, slow recall, or difficulty handling multiple pieces of information. By continuously adapting activities based on the child's strengths and weaknesses, the training becomes more engaging, supportive, and effective.

**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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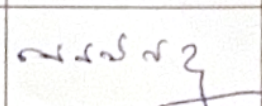
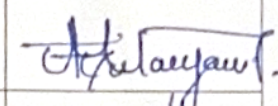
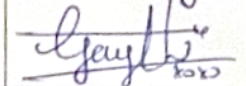
c) The student group can successfully execute the proposed project.

Yes	☒	No	☐
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d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Bhagyanie	Chathurika	
Co-Supervisor	Dr.	Kaushalya	Dissanayake	
External Supervisor	Ms.	Kanishka	Gayathri	

Summary of external supervisor's (if any) experience and expertise

I have worked as a child psychologist for over 10 years, specializing in supporting children with special learning disorders like dyslexia, dysgraphia and ADHD. I've conducted assessments, developed individualized plans and collaborated with parents and teachers to create inclusive learning environments. I would be very happy to help this team with their assignment.